

CSSE304 Exam 1 Paper Part

Please complete the following problems. If you have questions feel free to come up and ask (AJ), come to my classroom hours after this, or shoot me a message (Teams or Email works)! I will commit the solutions to this GitHub repo later tonight so you can review over the weekend. Good luck!

Problem 1

Write a function `sum-of-squared-odds` that takes in a list of integers and returns the sum of the squares of only the odd values. Use some combination of `map`, `apply`, and `filter` to complete the problem (no explicit recursion).

Ex. (`sum-of-squared-odds '(1 2 3 4 5)`) -> 35 ($1^2 + 3^2 + 5^2 = 35$)

Problem 2

Write a function `sum-of-string-lengths` that takes in a list of values (strings and numbers) and returns the sum of the lengths of only the strings. Use some combination of `map`, `apply`, and `filter` to complete the problem (no explicit recursion).

Ex. (`sum-of-string-lengths '("cat" 5 "a" 9 "tree")`) -> 8 (`length("cat") = 3`, `length("a") = 1`, `length("tree") = 4`) ($3 + 1 + 4 = 8$)

Problem 3

Write a function `pair-twister` that takes two procedures as arguments and returns a new procedure. The returned procedure takes two lists of equal length and does the following:

1. Apply the first procedure to each pair of corresponding elements from the two lists.
2. Combine all of the resulting values into a single value using the second procedure.

Ex. 1:

```
(define add-and-sum (pair-twister + +))
(add-and-sum '(1 2 3) '(10 20 30))
;; -> (apply + (map + '(1 2 3) '(10 20 30)))
;; -> (apply + '(11 22 33))
;; -> 66
```

Ex. 2:

```
(define mult-and-max (pair-twister * max))
(mult-and-max '(1 2 3) '(5 2 1))
;; -> (apply max (map * '(1 2 3) '(5 2 1)))
;; -> (apply max '(5 4 3))
;; -> 5
```

Hint: think about `map` and `apply`

Problem 4

Here is some code that uses `let` and `lambda` to show closures. Write what will be displayed by the code when it is run.

```
(define bank
  (let ((vault 500))
    (lambda ()
      (let ((withdrawals 0))
        (lambda ()
          (let ((fee 3))
            (set! vault (- vault 20))
            (set! withdrawals (+ withdrawals 1))
            (set! fee (+ fee withdrawals))
            (display (list vault withdrawals fee))
            (newline)))))))

(define teller1 (bank))
(define teller2 (bank))
(teller1)
(teller1)
(teller2)
(teller1)
```

Problem 5

Use the following grammar and expression to create a grammar tree. Use E for expr, B for base val, S for symbol, and I for integer.

```
<expr> ::= (<symbol> <expr> <expr>) | [<expr> <expr>] | <base val>
```

```
<base val> ::= <integer> | <symbol>
```

Problem 6

Consider the following grammar:

```
<expr> ::= <num> | (<expr> <op> <expr>)
<op> ::= + | - | * | /
<num> ::= <integer>
```

Write "yes" next to any of the following strings that are in this grammar, write "no" otherwise:

String	Yes/No
3	_____
(3 + 4)	_____
(3 + 4 * 5)	_____
((3 + 4) * 5)	_____
(3 + (4 * 5))	_____
(+ 3 4)	_____
(3 +)	_____
3 + 4	_____
((3 + 4) * (5 - 6))	_____
(3 ++ 4)	_____